Mathematics Standard
CBSE Examination Paper – 2023

With Topper's Answer

Time : 3:00 Hours **Max. Marks : 80**

General Instructions:

Read the following instructions carefully and follow them:

(i) *This question paper contains **38** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **FIVE** sections - **A, B, C, D** and **E.***

(iii) *In **Section A,** Question numbers **1** to **18** are multiple choice questions (MCQs) and question numbers **19** and **20** are Assertion – Reason based questions of **1** mark each.*

(iv) *In **Section B,** Question numbers **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C,** Question numbers **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D,** Question numbers **32** to **35** are long answer (LA) type questions, carrying **5** marks each.*

(vii) *In **Section E,** Question numbers **36** to **38** are **case-study based integrated** questions carrying **4** marks each. Internal choice is provided in **2** marks questions in each case-study.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section **B**, 2 questions in Section **C**, 2 questions in Section **D** and 3 questions of 2 marks in Section **E**.*

(ix) *Draw neat diagrams wherever required. Take $\pi = \frac{22}{7}$ wherever required, if not stated.*

(x) *Use of calculators is **NOT allowed**.*

Set--30/5/1

SECTION – A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. The number of polynomials having zeroes -3 and 5 is :

(a) only one **(b)** infinite **(c)** exactly two **(d)** at most two

Ans. (b) infinite

2. The pair of equations $ax + 2y = 9$ and $3x + by = 18$ represent parallel lines, where a, b are integers, if :

(a) $a = b$ **(b)** $3a = 2b$ **(c)** $2a = 3b$ **(d)** $ab = 6$

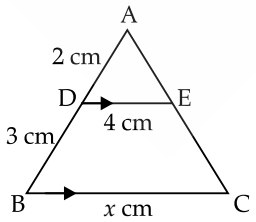
Ans. (d) $ab = 6$

3. The common difference of the A.P. whose nth term is given by $a_n = 3n + 7$, is :

(a) 7 **(b)** 3 **(c)** $3n$ **(d)** 1

Ans. (b) 3

4. In the given figure, DE || BC. The value of x is:



(a) 6 **(b)** 12.5 **(c)** 8 **(d)** 10

Ans. (d) 10

5. A quadratic equation whose roots are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$ is :

(a) $x^2 - 4x + 1 = 0$ **(b)** $x^2 + 4x + 1 = 0$ **(c)** $4x^2 - 3 = 0$ **(d)** $x^2 - 1 = 0$

Ans. (a) $x^2 - 4x + 1 = 0$

6. If $\tan\theta = \frac{5}{12}$, then the value of $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta}$ is :

(a) $-\frac{17}{7}$ **(b)** $\frac{17}{7}$ **(c)** $\frac{17}{13}$ **(d)** $-\frac{7}{13}$

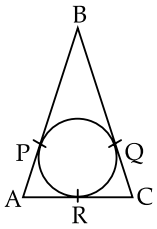
Ans. (a) $-\frac{17}{7}$

7. The distance between the points $P\left(-\frac{11}{3}, 5\right)$ and $Q\left(-\frac{2}{3}, 5\right)$ is :

(a) 6 units **(b)** 4 units **(c)** 2 units **(d)** 3 units

Ans. (d) 3 units

8. In the given figure, AB = BC = 10 cm. If AC = 7 cm, then the length of BP is :



(a) 3.5 cm **(b)** 7 cm **(c)** 6.5 cm **(d)** 5 cm

Ans. (c) 6.5 cm

9. Water in a river which is 3 m deep and 40 m wide is flowing at the rate of 2 km/h. How much water will fall into the sea in 2 minutes ?

(a) 800 m^3 **(b)** 4000 m^3 **(c)** 8000 m^3 **(d)** 2000 m^3

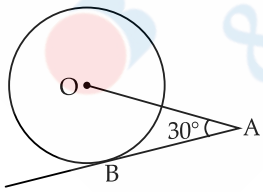
Ans. (c) 8000 m^3

10. If the mean and the median of a data are 12 and 15 respectively, then its mode is:

(a) 13.5 **(b)** 21 **(c)** 6 **(d)** 14

Ans. (b) 21

11. In the given figure, AB is a tangent to the circle centered at O. If OA = 6 cm and $\angle OAB = 30°$, then the radius of the circle is:



(a) 3 cm **(b)** $3\sqrt{3}$ cm **(c)** 2 cm **(d)** $\sqrt{3}$ cm

Ans. (a) 3 cm

12. $\left(\frac{2\tan 30°}{1+\tan^2 30°}\right)$ is equal to :

(a) sin 60° **(b)** cos 60° **(c)** tan 60° **(d)** sin 30°

Ans. (a) sin 60°

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13. In Δ ABC and Δ DEF, $\frac{AB}{DE}=\frac{BC}{FD}$. Which of the following makes the two triangles similar ?

(a) $\angle A = \angle D$ **(b)** $\angle B = \angle D$ **(c)** $\angle B = \angle E$ **(d)** $\angle A = \angle F$

Ans. (b) $\angle B = \angle D$

14. The 11th term from the end of the A.P. : 10, 7, 4, ... ,– 62 is :

(a) 25 **(b)** 16 **(c)** – 32 **(d)** 0

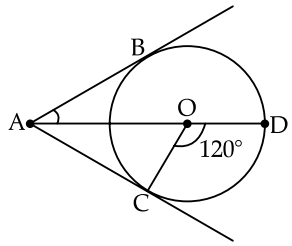
Ans. (c) – 32

15. Two coins are tossed together. The probability of getting at least one tail is :

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** 1

Ans. (c) $\frac{3}{4}$

16. In the given figure, AC and AB are tangents to a circle centered at O. If $\angle COD = 120°$, then $\angle BAO$ is equal to:



(a) 30° **(b)** 60° **(c)** 45° **(d)** 90°

Ans. (a) 30°

17. Which of the following numbers cannot be the probability of happening of an event?

(a) 0 **(b)** $\frac{7}{0.01}$ **(c)** 0.07 **(d)** $\frac{0.07}{3}$

Ans. (b) $\frac{7}{0.01}$

18. If every term of the statistical data consisting of n terms is decreased by 2, then the mean of the data :

(a) decreases by 2 **(b)** remains unchanged

(c) decreases by $2n$ **(d)** decreases by 1

Ans. (a) decreases by 2

Questions number **19** and **20** are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled as Assertion (a) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion (a) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (a).

(b) Both Assertion (a) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (a).

(c) Assertion (a) is true, but Reason (R) is false.

(d) Assertion (a) is false, but Reason (R) is true.

19. Assertion (a): If the points A(4, 3) and B(x, 5) lie on a circle with centre O(2, 3), then the value of x is 2.
Reason (R): Centre of a circle is the mid-point of each chord of the circle.

Ans (c) Assertion (A) is true, but Reason (R) is false.

20. Assertion (a): The number 5^n cannot end with the digit 0, where n is a natural number.
Reason (R): Prime factorisation of 5 has only two factors, 1 and 5.

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Ans. (c) Assertion (A) is true, but Reason (R) is false.
[only 1 prime factor = 5 ∵ prime factorisation of a prime number is the number itself]

SECTION – B

This section comprises very short answer (VSA) type questions of 2 marks each.

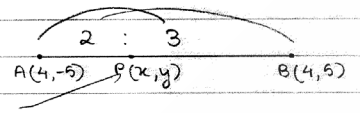
21. (a) The line segment joining the points A(4,–5) and B(4, 5) is divided by the point P such that AP : AB = 2 : 5. Find the coordinates of P.

OR

(b) Point P(x, y) is equidistant from points A(5, 1) and B(1, 5). Prove that $x = y$.

Ans. A(4, −5) and B(4, 5)

(a) Let the coordinates of point P which divides AB such that AP : AB = 2 : 5 be P(x, y).



$\frac{AP}{AB} = \frac{2}{5}$

$\Rightarrow \frac{AP + PB}{AP} = \frac{5}{2}$

$\Rightarrow 2PB = 3AP \Rightarrow AP : PB = 2 : 3$

P divides AB internally in the ratio of 2 : 3

By section formula,

$P(x, y) = P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

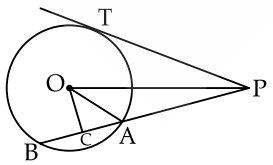
$\Rightarrow P(x, y) = P\left(\frac{2\times4 + 3\times4}{2+3}, \frac{2\times5 + 3\times(-5)}{2+3}\right)$

$\Rightarrow P(x, y) = P\left(\frac{20}{5}, \frac{-5}{5}\right) = P(4, -1)$

∴ Coordinates of P are P(4, −1).

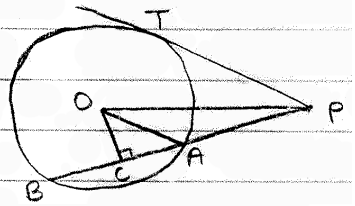
22. In the given figure, PT is a tangent to the circle centered at O. OC is perpendicular to chord AB. Prove that PA.

PB = $PC^2 - AC^2$.



Ans. Given: Circle with centre O
PT is a tangent
OC ⊥ AB

To Prove: $PA \cdot PB = PC^2 - AC^2$



Perpendicular from centre to chord bisects it
$\Rightarrow$ C is midpoint of AB ($\because \angle OCA = 90°$, given)
$\Rightarrow BC = CA = \frac{AB}{2}$

$\Rightarrow AB = 2AC$

By Pythagoras Theorem,
In right $\triangle OCA$, $OA^2 = OC^2 + AC^2$
In right $\triangle OCP$, $OP^2 = OC^2 + PC^2$

LHS = PA · PB
= PA × (PA + AB)
= $PA^2 + PA \cdot AB$
= $PA^2 + PA \cdot 2AC$ (from ①)
= $(PA)^2 + 2(PA)(AC)$
= $(PA + AC)^2 - AC^2$ [$(a+b)^2 = a^2 + 2ab + b^2$]
= $PC^2 - AC^2$ = RHS
Hence, proved

23. Using prime factorisation, find HCF and LCM of 96 and 120.

Ans. First number = 96
Second number = 120

$96 = 2^5 \times 3$
$120 = 2^3 \times 3 \times 5$
HCF (96, 120) = $2^3 \times 3$
= 8 × 3
= 24

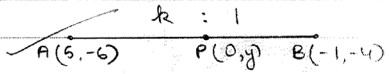
2 | 96	2 | 120
2 | 48	2 | 60
2 | 24	2 | 30
2 | 12	3 | 15
2 | 6	5
3	

LCM (96, 120) = $2^5 \times 3 \times 5$
= 32 × 3 × 5
= 480

$\therefore$ HCF (96, 120) = 24 and LCM (96, 120) = 480

24. Find the ratio in which y-axis divides the line segment joining the points (5, – 6) and (– 1, – 4).

Ans. Points are A(5, -6) and B(-1, -4)
Let the point where y-axis ($x = 0$) intersects AB be P(0, y)



Let the ratio in which P(0, y) divides AB be k : 1

By section formula,

$$P(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

$$P(0, y) = \left(\frac{-k+5}{k+1}, \frac{-4k+(-6)}{k+1}\right)$$

$0 = \frac{-k+5}{k+1}$ and $y = \frac{-4k-6}{k+1}$

$0 = -k + 5$

$k = 5$

$k : 1 = 5 : 1$

$\therefore$ Ratio is $\boxed{5:1}$

25. (a) If $a \cos\theta + b \sin\theta = m$ and $a \sin\theta - b \cos\theta = n$, then prove that $a^2 + b^2 = m^2 + n^2$.

OR

(b) Prove that :

$$\sqrt{\frac{\sec A - 1}{\sec A + 1}} + \sqrt{\frac{\sec A + 1}{\sec A - 1}} = 2\text{coses A}$$

Ans.

(a) $a\cos\theta + b\sin\theta = m$

$a\sin\theta - b\cos\theta = n$

To prove: $a^2 + b^2 = m^2 + n^2$

$a\cos\theta + b\sin\theta = m$

Squaring both sides,

$a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta = m^2$ $\left[(a+b)^2 = a^2 + b^2 + 2ab\right]$

$a\sin\theta - b\cos\theta = n$

Squaring both sides,

$a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta = n^2$ $\left[(a-b)^2 = a^2 + b^2 - 2ab\right]$

Adding ① and ②,

$(a^2\cos^2\theta + a^2\sin^2\theta) + (b^2\sin^2\theta + b^2\cos^2\theta) + 2ab\sin\theta\cos\theta - 2ab\sin\theta\cos\theta = m^2 + n^2$

$\Rightarrow a^2(\sin^2\theta + \cos^2\theta) + b^2(\sin^2\theta + \cos^2\theta) = m^2 + n^2$

$\because \sin^2\theta + \cos^2\theta = 1$

$\Rightarrow a^2 + b^2 = m^2 + n^2$

LHS = RHS

Hence, proved

SECTION – C

This section comprises short answer (SA) type questions of 3 marks each.

26. (a) Prove that $\sqrt{3}$ is an irrational number.

OR

(b) The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they change simultaneously at 7 a.m., at what time will they change together next ?

Ans.

(a) Let us assume, to the contrary, that $\sqrt{3}$ is rational.

$\Rightarrow \sqrt{3} = \frac{p}{q}$ where $q \neq 0$, p and q are coprime positive integers

Squaring both sides,

$\Rightarrow 3 = \frac{p^2}{q^2}$

$\Rightarrow p^2 = 3q^2$

$\Rightarrow$ 3 divides p^2

$\Rightarrow$ 3 divides p ($\because$ 3 is prime)

So, let $p = 3m$

Substituting in

$(3m)^2 = 3q^2$

$\Rightarrow 9m^2 = 3q^2$

$\Rightarrow q^2 = 3m^2$

$\Rightarrow$ 3 divides q^2

$\Rightarrow$ 3 divides q

From ③ and ④,

3 divides both p and q

But p and q are coprime, i.e. HCF$(p, q) = 1$ (Using ①)

which is a contradiction

$\therefore$ our supposition is wrong

$\therefore \sqrt{3}$ must be irrational.

Hence, proved

27. If p^{th} term of an A.P. is q and q^{th} term is p, then prove that its nth term is $(p + q - n)$.

Ans. Let the first term of AP be a and common difference be d

AP: $a, a+d, a+2d, \ldots$

$a_n = a + (n-1)d$

$a_p = a + (p-1)d = q$

$\Rightarrow a + pd - d = q$ ①

$a_q = a + (q-1)d = p$

$\Rightarrow a + qd - d = p$ ②

Subtracting ① from ②,

$a + qd - d = p$

$\ominus \quad a + pd - d = q$

$(q-p)d = p-q$

$\Rightarrow d = -1$

Substituting in ①,

$a + p(-1) - (-1) = q$

$\Rightarrow a + 1 = p + q$

$a_n = a + (n-1)d$

$= a + (n-1)(-1)$

$= a - n + 1$

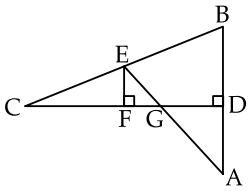
$= (a+1) - n$

$= p + q - n$ (Substituting from (3))

LHS = RHS

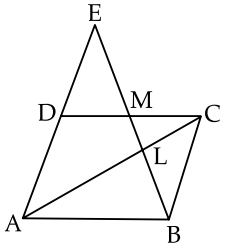
Hence, proved

28. (a) In the given figure, CD is the perpendicular bisector of AB. EF is perpendicular to CD. AE intersects CD at G. Prove that $\frac{CF}{CD} = \frac{FG}{DG}$.



OR

(b) In the given figure, ABCD is a parallelogram. BE bisects CD at M and intersects AC at L. Prove that EL = 2BL.

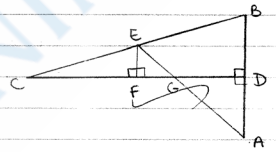


Ans.

(a) Given: CD is ⊥ bisector of AB

EF ⊥ CD

AE intersects CD at G



To Prove: $\frac{CF}{CD} = \frac{FG}{DG}$

Proof: In △EFG and △ADG,

∠EFG = ∠ADG = 90° (given, linear pair with ∠EFC and ∠BDG)

∠EGF = ∠AGD (vertically opposite angles)

∴ △EFG ~ △ADG (by AA similarity criterion)

$\Rightarrow \frac{EF}{AD} = \frac{FG}{DG}$ (cpst)

In △ECF and △BCD,

∠EFC = ∠BDC = 90° (given)

∠ECF = ∠BCD (common angle)

∴ △ECF ~ △BCD (by AA similarity criterion)

$\Rightarrow \frac{EF}{BD} = \frac{CF}{CD}$ (cpst)

But AD = BD (∵ CD bisects AB)

$\Rightarrow \frac{EF}{AD} = \frac{EF}{BD} = \frac{CF}{CD}$

From item ① and ②,

$$\frac{EF}{AD} = \frac{CF}{CD} = \frac{FG}{DG}$$

$$\therefore \frac{FG}{CD} \frac{CF}{DG} = \frac{FG}{DG}$$

Hence, proved

29. Two people are 16 km apart on a straight road. They start walking at the same time. If they walk towards each other with different speeds, they will meet in 2 hours. Had they walked in the same direction with same speeds as before, they would have met in 8 hours. Find their walking speeds.

Ans. Let the speed of person 1 be x km/h and speed of person 2 be y km/h $(x > y)$

distance = speed × time

Given distance = 16 km

Case 1 : Towards each other

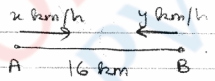
Time = 2 h

Speed = $(x+y)$ km/h

Distance = $2(x+y)$ km

$\Rightarrow 16 = 2x + 2y$

$\Rightarrow x + y = 8$ ...①



Case 2 Same direction

Time = 8 h

Speed = $(x-y)$ km/h

Distance (only AB) = $8(x-y)$ $[16 + 8y = 8x]$

$\Rightarrow 16 = 8x - 8y$

$\Rightarrow x - y = 2$ ...②



Adding ① and ②,

$$x + y = 8$$
$$+ \quad x - y = 2$$
$$2x = 10$$
$$\Rightarrow x = 5$$

Substituting in ①,

$y = 3$ km

$\therefore$ Speed of person 1 = 5 km/h

Speed of person 2 = 3 km/h

30. Prove that :

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$$

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Ans. $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$

$$\text{LHS} = \frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta}$$

$$= \frac{\tan\theta}{1-\frac{1}{\tan\theta}} + \frac{\frac{1}{\tan\theta}}{1-\tan\theta}$$

$$= \frac{\tan^2\theta}{\tan\theta - 1} + \frac{1}{\tan\theta(1-\tan\theta)}$$

$$= \frac{\tan^2\theta}{\tan\theta - 1} - \frac{1}{\tan\theta(\tan\theta - 1)}$$

$$= \frac{\tan^3\theta - 1}{(\tan\theta - 1)\tan\theta}$$

$$= \frac{(\tan\theta - 1)(\tan^2\theta + 1 + \tan\theta)}{\tan\theta(\tan\theta - 1)} \qquad \left[\frac{(a-b)(a^2+b^2+ab)}{a^3-b^3}\right]$$

$$= \frac{1 + \tan^2\theta + \tan\theta}{\tan\theta}$$

$$= \frac{\tan\theta + \sec^2\theta}{\tan\theta} \qquad (1 + \tan^2\theta = \sec^2\theta)$$

$$= \frac{\tan\theta}{\tan\theta} + \frac{\sec^2\theta}{\tan\theta}$$

$$= 1 + \frac{1}{\cos^2\theta} \times \frac{\cos\theta}{\sin\theta} \qquad \left(\sec\theta = \frac{1}{\cos\theta},\ \tan\theta = \frac{\sin\theta}{\cos\theta}\right)$$

$$= 1 + \frac{1}{\cos\theta\sin\theta}$$

$$= 1 + \sec\theta\,\text{cosec}\,\theta$$

$$= \text{RHS}$$

Hence, proved

31. Find the mean of the following frequency distribution:

Classes	25-30	30-35	35-40	40-45	45-50	50-55	55-60
Frequency	14	22	16	6	5	3	4

Ans.

Classes	Frequency (f_i)	Midpoints (x_i)	$u_i = \frac{x_i - A}{h}$	$f_i u_i$
25 - 30	14	27.5	-3	-42
30 - 35	22	32.5	-2	-44
35 - 40	16	37.5	-1	-16
40 - 45	6	A (42.5)	0	0
45 - 50	5	47.5	1	5
50 - 55	3	52.5	2	6
55 - 60	4	57.5	3	12
	$\Sigma f_i = 70$			$\Sigma f_i u_i = -79$

Mean = $\bar{x} = A + \frac{h\Sigma f_i u_i}{\Sigma f_i}$

$= 42.5 + \frac{5 \times (-79)}{70}$

$= 42.5 - \frac{395}{70}$

$= 42.5 - 5.642$

$= 36.858 \approx 36.86$

∴ Mean = 36.86 (approx)

SECTION – D

This section comprises long answer (LA) type questions of 5 marks each.

32. One observer estimates the angle of elevation to the basket of a hot air balloon to be 60°, while another observer 100m away estimates the angle of elevation to be 30°. Find :

(a) The height of the basket from the ground.

(b) The distance of the basket from the first observer's eye.

(c) The horizontal distance of the second observer from the basket.

Ans.

Introduction

2 cases are formed

A = hot-air ballon

C = first observer

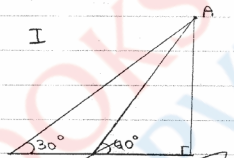
D = second observer

Angle of elevation of A from C = ∠ACB = 60°

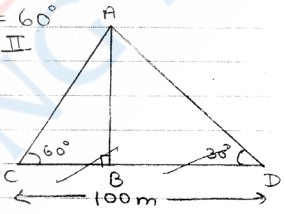
Angle of elevation of A from BD = ∠ADB = 30°

CD = 100 m

I



II



(a) Let height of basket = AB = h m

Case I : In △ABC, $\tan 60° = \frac{AB}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{BC}$

$\Rightarrow BC = \frac{h}{\sqrt{3}}$ (1)

In △ABD, $\tan 30° = \frac{AB}{BD}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BC + 100}$

$\Rightarrow BC + 100 = h\sqrt{3}$

$BC + 100 = h\sqrt{3}$

$\Rightarrow \frac{h}{\sqrt{3}} + 100 = h\sqrt{3}$

$\Rightarrow 100 = h\sqrt{3} - \frac{h\sqrt{3}}{3}$

$\Rightarrow \frac{2h\sqrt{3}}{3} = 100$

$\Rightarrow h = \frac{150}{\sqrt{3}} = 50\sqrt{3}$ m

∴ Height of basket = $\boxed{50\sqrt{3} \text{ m}}$

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Case 2 : In $\triangle ABC$,

$\tan 60^\circ = \frac{AB}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{BC}$

$\Rightarrow BC = \frac{h}{\sqrt{3}}$

In $\triangle ABD$,

$\tan 30^\circ = \frac{AB}{BD}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BD}$

$\Rightarrow BD = h\sqrt{3}$

$BC + BD = 100\ m$

$\Rightarrow \frac{h}{\sqrt{3}} + h\sqrt{3} = 100$

$\Rightarrow \frac{4h\sqrt{3}}{3} = 100$

$\Rightarrow h = 25\sqrt{3}$

$\therefore$ Height of basket = $\boxed{25\sqrt{3}\ m}$

(b). Case 1 : Dist of A from C = AC

In $\triangle ABC$,

$\sin 60^\circ = \frac{AB}{AC}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{50\sqrt{3}}{AC}$

$\Rightarrow AC = 100\ m$

$\therefore$ Distance of basket from first observer = $\boxed{100\ m}$

Case 2 : Dist. of A from C = AC

In $\triangle ABC$,

$\sin 60^\circ = \frac{AB}{AC}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{25\sqrt{3}}{AC}$

$\Rightarrow AC = 50\ m$

$\therefore$ Distance of basket from first observer = $\boxed{50\ m}$

(c) Case 1 : To find - BD

$BD = BC + CD$

$= \frac{h}{\sqrt{3}} + 100$ (from ①)

$= \frac{50\sqrt{3}}{\sqrt{3}} + 100$

$= 50 + 100 = 150$

$\therefore$ Horizontal distance BD = $\boxed{150\ m}$

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Case 2 : To find – BD

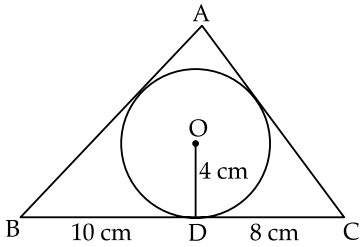
$BD = h\sqrt{3}$ (from (2))

$= 25\sqrt{3} \times \sqrt{3}$

$= 75$

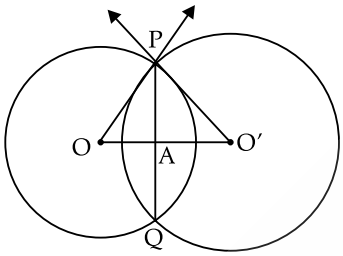
∴ Horizontal distance BD = 75 m

33. (a) A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC are of lengths 10 cm and 8 cm respectively. Find the lengths of the sides AB and AC, if it is given that area Δ ABC = 90 cm^2.



OR

(b) Two circles with centres O and O' of radii 6 cm and 8 cm, respectively intersect at two points P and Q such that OP and O'P are tangents to the two circles. Find the length of the common chord PQ.



Ans.

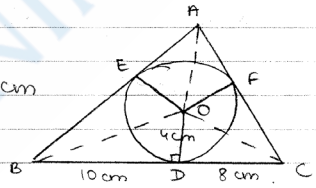
(a) Given : △ABC

Incircle with radius $r = 4$ cm

BD = 10 cm

CD = 8 cm

ar (ABC) = 90 cm^2



To find : Lengths of AB and AC.

Construction : Join OE, OF, OA, OB, OC

OD = OE = OF = r = 4cm (radii)

Tangents from same external point are equal in length.

From pt. A, AE = AF = x (let it be)

From pt. B, BE = BD = 10 cm

From pt. C, CD = CF = 8 cm

Area of $\Delta = \frac{1}{2} \times b \times h$

Area of △ABC = Area of △OAB + Area of △OBC + Area of △OCA

$\Rightarrow 90 = \frac{1}{2} \times OE \times AB + \frac{1}{2} \times OD \times BC + \frac{1}{2} \times OF \times AC$

$$+\frac{1}{2}\times x\times(CF+AF)$$

$$\Rightarrow \quad 90=\frac{1}{2}\times x\times(x+10)+\frac{1}{2}\times x\times(10+8)+\frac{1}{2}\times x\times(8+x)$$

$$\Rightarrow \quad 90=\frac{1}{2}\times 4\,(x+10+10+8+8+x)$$

$$\Rightarrow \quad 45=2x+36$$

$$\Rightarrow \quad 2x=9$$

$$\Rightarrow \quad x=4.5$$

$$AB=AE+BE=4.5+10=14.5\text{ cm}$$

$$AC=AF+CF=4.5+8=12.5\text{ cm}$$

$$\therefore \quad AB=14.5\text{ cm}$$

$$AC=12.5\text{ cm}$$

34. (a) A train travels at a certain average speed for a distance of 54 km and then travels a distance of 63 km at an average speed of 6 km/h more than the first speed. If it takes 3 hours to complete the journey, what was its first average speed ?

OR

(b) Two pipes together can fill a tank in $\frac{15}{8}$ hours. The pipe with 8 larger diameter takes 2 hours less than the pipe with smaller diameter to fill the tank separately. Find the time in which each pipe can fill the tank separately.

Ans. (a) Let the original average speed be x km/h.

Original : distance = 54 km

speed = x km/h

$$\text{time}=\frac{\text{distance}}{\text{speed}}=\frac{54}{x}\text{ h}$$

New : distance = 63 km

speed = $(x+6)$ km/h

$$\text{time}=\frac{\text{distance}}{\text{speed}}=\frac{63}{x+6}\text{ h}$$

$$\frac{54}{x}+\frac{63}{x+6}=3$$

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$$\Rightarrow 3\left(\frac{18}{x}+\frac{21}{x+6}\right)=3$$

$$\Rightarrow \frac{18x+108+21x}{x^2+6x}=1$$

$$\Rightarrow 39x+108=x^2+6x$$

$$\Rightarrow x^2-33x-108=0$$

$$\Rightarrow x^2-36x+3x-108=0$$

$$\Rightarrow x(x-36)+3(x-36)=0$$

$$\Rightarrow (x-36)(x+3)=0$$

$$\Rightarrow x=36 \text{ or } x=-3$$

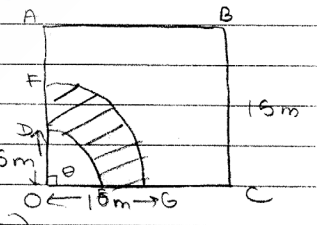
∵ speed cannot be negative,
$x=-3$ will be neglected.

$\Rightarrow x=36$

∴ Average speed of train (original) = **36 km/h**

35. A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze. Also, find the increase in grazing area if length of rope is increased to 10m. (Use $\pi = 3.14$)

Ans.



Square field OABC
Side = s = 15m

Quadrant
$\angle DOE = \theta = 90^\circ$ (all angles of square $= 90^\circ$)
Radius = length of rope
$= r = 5$ m

Area grazed by horse = Area of sector DOE

$$= \frac{\theta}{360^\circ}\times \pi \times r^2$$

$$= \frac{90}{360}\times \frac{157}{50}\times 5\times 5$$

$$= \frac{3925}{200}$$

$$= 19.625 \text{ m}^2$$

157
25
785
314×
3925
2 7775
= 1962 5

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New, new radius = new length of rope = R = 10 m

Increase in grazing area = Area of sector FOG − Area of sector DOE

$$= \frac{\theta}{360^\circ} \times \pi \times R^2 - \frac{\theta}{360^\circ} \times \pi \times r^2$$

$$= \frac{90}{360} \times \frac{314}{100} \times (10^2 - 5^2)$$

$$= \frac{90}{360} \times \frac{157}{100} \times 15 \times 5 \qquad [a^2 - b^2 = (a+b)(a-b)]$$

$$= \frac{11775}{200}$$

$$= 58.875 \text{ m}^2$$

Original area grazed = 1.9625 m²

Increase in area = 58.875 m²

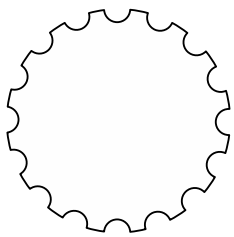
SECTION – E

This section comprises 3 case study based questions of 4 marks each.

Case Study - 1

36. A golf ball is spherical with about 300 - 500 dimples that help increase its velocity while in play. Golf balls are traditionally white but available in colours also. In the given figure, a golf ball has diameter 4.2 cm and the surface has 315 dimples (hemi-spherical) of radius 2 mm.





Based on the above, answer the following questions :

(i) Find the surface area of one such dimple. **1**

(ii) Find the volume of the material dug out to make one dimple. **1**

(iii)(a) Find the total surface area exposed to the surroundings. **2**

OR

(iii)(b) Find the volume of the golf ball. **2**

Ans.

Golf ball = Sphere

Radius = $R = \frac{4.2}{2} = 2.1$ cm

Dimple = Hemisphere

Radius = $r = 2$ mm $= 0.2$ cm

(i) SA of 1 dimple = CSA of hemisphere

$= 2\pi r^2$

$= 2 \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10}$

$= \frac{176}{700}$ cm^2

$= 0.2514$ cm^2

(ii) Vol to make 1 dimple = Vol of hemisphere

$= \frac{2\pi r^3}{3}$

$= \frac{2}{3} \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10} \times \frac{2}{10}$

$= \frac{352}{21000}$

$= 0.01676$ cm^3

(iii) (iv) Vol of golf ball = Vol of sphere − Vol of 315 hemispheres

$= \frac{4\pi R^3}{3} - \frac{2\pi r^3 \times 315}{3}$

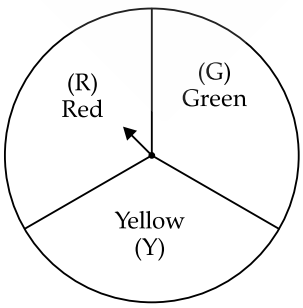
$= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{10} \times \frac{21}{10} \times \frac{21}{10} - \frac{2}{3} \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10} \times 315$

$= 38.808 - 5.28$

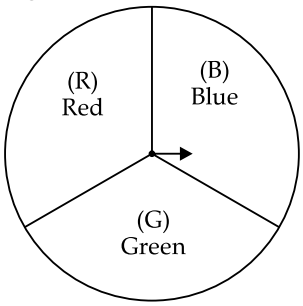
$= 33.588$ ~~33.588~~ 33.528 cm^3

Case Study - 2

37. A middle school decided to run the following spinner game as a fund-raiser on Christmas Carnival.



I



II

Making Purple : Spin each spinner once. Blue and red make purple. So, if one spinner shows Red (R) and another Blue (b), then you 'win'. One such outcome is written as 'RB'.

Based on the above, answer the following questions :

(i) List all possible outcomes of the game. **1**

(ii) Find the probability of 'Making Purple'. **1**

(iii)(a) For each win, a participant gets ₹ 10, but if he/she loses, he/she has to pay ₹ 5 to the school. If 99 participants played, calculate how much fund could the school have collected. **2**

OR

(iii)(b) If the same amount of ₹ 5 has been decided for winning or losing the game, then how much fund had been collected by school? (Number of participants = 99) **2**

Ans.

(i) Spinner I – Spinner II

Red (R) – Red (R) { RR,
Red (R) – Blue (B) RB,
Red (R) – Green (G) RG,
Green (G) – Red (R) GR,
Green (G) – Blue (B) GB,
Green (G) – Green (G) GG,
Yellow (Y) – Red (R) YR,
Yellow (Y) – Blue (B) YB,
Yellow (Y) – Green (G) YG }

Total no. of outcomes = 9

(ii) X = {RB}

Favourable outcomes = 1

$P(\text{making purple}) = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}} = \frac{1}{9}$

(iii)(a) No. of participants = 99

Winning

No. $= \frac{1}{9} \times 99 = 11$

Amount = ₹(−10) (school has to pay)

Value = ₹(−110) [loss]

Loss

No. = 99 − 11 = 88

Amount = ₹5 (pay to school)

Value = ₹440 [gain]

Net = +440 − 110

= +330

∴ So, the school most likely collected Rs. 330

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Case Study - 3

38. In a pool at an aquarium, a dolphin jumps out of the water travelling at 20 cm per second. Its height above water level after t seconds is given by $h = 20t - 16t^2$.

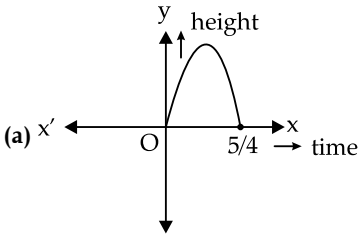


Based on the above, answer the following questions :

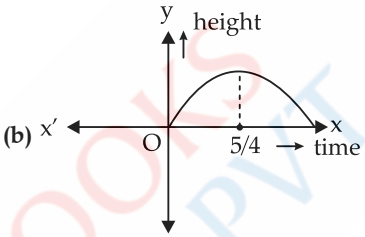
(i) Find zeroes of polynomial $p(t) = 20t - 16t^2$. **1**

(ii) Which of the following types of graph represents $p(t)$? **1**

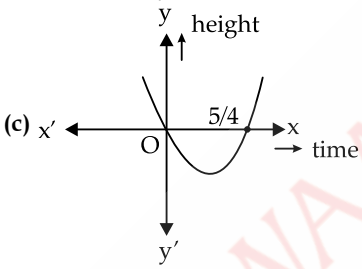
(a)



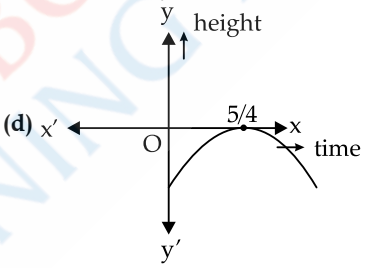
(b)



(c)



(d)



Ans.

(iii)(a) What would be the value of h at $t = \frac{3}{2}$? Interpret the result. **2**

OR

(b) How much distance has the dolphin covered before hitting the water level again ? 2

(i) $p(t) = 20t - 16t^2$

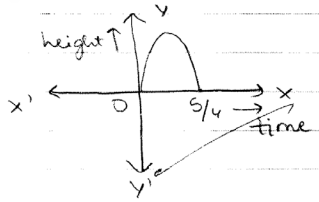
$20t - 16t^2 = 0$

$-4t(4t - 5) = 0$

$t = 0$ or $t = \frac{5}{4}$

∴ Zeroes of $p(t) = 20t - 16t^2$ are 0 and $\frac{5}{4}$

(ii) (a)



→ opens downwards ($a < 0$)

→ intersects x-axis at $(0, 0)$ and $\left(\frac{5}{4}, 0\right)$

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(iii) Water level is hit when $h = 0$

(b) $h = 20t - 16t^2$

$\Rightarrow 0 = -4t(4t - 5)$

$\Rightarrow t = 0$ or $t = \frac{5}{4}$

$\because$ Delphin has started at $t = 0$

$\Rightarrow$ at $t = \frac{5}{4}$, delphin reaches water level again.

distance covered = speed × time

$= 20 \text{ cm/s} \times \frac{5}{4} \text{ s}$ 2

$= \boxed{25 \text{ cm}}$

■■

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